

Research Article

Real-Time Automatic Detection of Nutrient Deficiency in Lettuce Plants With New YOLOV12 model

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Healthy lettuce receives all the necessary nutrients, such as calcium, magnesium, nitrogen, phosphorus, and potassium for optimum growth. Deficiency of one of these vitamins causes growth retardation in plants. Vitamin deficiencies in lettuce grown in hydroponic systems can negatively affect plant development and create various problems. Therefore, it is critical to provide all the vitamin and mineral balances that plants need in hydroponic systems. In this study, the development of lettuce, which has an important place in daily life, was examined in detail. Deep learning techniques were applied to images of lettuce plants using various You Only Look Once (YOLO) models (YOLOV12 and YOLOV11). Various metrics were calculated for each YOLO version to evaluate performance. These metrics include precision, recall, mean average precision (mAP)50 (average accuracy), and mAP50–95 (average accuracy at different intersection over union [IoU]), thresholds). A balance was created between the classes on the lettuce images obtained using data augmentation techniques, which allowed the model to be trained more robustly. In the tests conducted with the augmented dataset, the mAP50 value of the YOLOV12 model was calculated as 0.984 and the mAP50–95 value as 0.957. This reveals the high performance of the model in correctly recognizing nutritional deficiencies. On the other hand, the YOLOV11 model achieved lower results, such as 0.975 mAP50 and 0.944 mAP50–95 in all classes. This study significantly improved the object detection performance by integrating novel attention modules into YOLOV11 and YOLOV12 models to detect nutrient deficiencies of lettuce plants early and accurately in soilless agriculture. Not only the comparison of models, but also the generalization and robustness of the model were strengthened with innovative modules that provide dynamic adaptation to environmental variables. This novel approach provides tangible contributions to both academic literature and practical applications in AI-supported early detection and yield optimization in sustainable hydroponic agriculture.

Keywords: computer vision; mAP50; mAP50–95; nutrient deficiency; YOLO

1. Introduction

As the world population continues to grow rapidly in the 21st century, the challenges of food production are increasing. According to the United Nations, [1] the world population is expected to reach 9.7 billion by 2050. We need an increase in food production in order for current agricultural systems to feed the rapidly growing population [2]. Factors such as climate change, water scarcity, and soil degradation threaten the sustainability of traditional agriculture [3]. Innovative agricultural methods, such as soilless agriculture have great potential to

ensure future food security and accelerate food production [4]. Soilless agriculture is a method of agriculture in which plants are grown in mineral nutrient solutions instead of soil. Plants receive the nutrients they need from the solution, and the need to be attached to the soil is eliminated. The roots of soilless agriculture date back to ancient civilizations. The Hanging Gardens of Ancient Babylon and the agricultural areas built on islands by the Aztecs can be considered as examples in the past [5, 6]. The first soilless agriculture in the modern sense was introduced by Dr. Williams F. Gericke in 1929 and has become widespread rapidly since the mid-20th century [7].

Traditional agricultural methods are based on the management of soil, water, and nutrient resources used in agricultural production. However, these methods face several challenges, especially when compared to modern agricultural demands [8]. First, traditional methods are often based on human observation and experience. This can make it difficult to diagnose nutrient deficiencies accurately and timely [9]. Visual symptoms (e.g., leaf yellowing or deformities) usually appear in advanced stages of nutrient deficiency, which can lead to yield losses [10]. In addition, the accuracy of human observations decreases as the field size increases, making early detection of deficiencies even more difficult. Another challenge is the impact of environmental factors on nutrient management. Variables such as soil structure, weather, and irrigation can make it difficult to distribute nutrients evenly to plants [11]. Traditional methods cannot respond quickly and effectively to these dynamic variables. For example, a sudden change in soil pH can prevent the uptake of certain nutrients by the plant; this problem cannot be detected quickly with traditional approaches.

Soilless farming (hydroponic) systems have various subsystems depending on how water and nutrient solution are delivered to plant roots. The most commonly used systems are nutrient film technique (NFT), deep water culture (DWC), aeroponics, drip irrigation systems, and wick systems [12]. In the NFT system, a thin nutrient solution flows slowly under the roots and allows the roots to receive oxygen. In the DWC system, plants are suspended in a continuously oxygenated nutrient solution. In aeroponics systems, plant roots are fed with a nutrient solution suspended in the air and sprayed in the form of a fine mist [7, 13]. Each system has its own advantages and disadvantages. The system to be used is selected depending on the plant type, growing conditions, targeted product amount, and desired investment [14]. One of the main advantages of soilless farming is protection from pests and weeds caused by soilborne pathogens. This minimizes the use of pesticides and minimizes the damage to the environment [13]. Soilless agriculture also offers a sustainable agricultural solution in regions with low water resources by using less water than traditional agriculture and recycling used water resources [7]. The growth environment and nutrient content of plants can be completely controlled, which increases product quality and carries out the growth process in the most optimum way [15]. Hydroponic systems take up less space and enable urban agriculture with vertical farming applications. It helps to grow fresh products in cities and distribute them quickly. Thus, it helps to meet the fresh product needs of urban residents [16]. Soilless agriculture offers not only environmental but also economic advantages. It offers high efficiency when compared to traditional farming methods. It allows for vertical farming by placing more products on the same area, which helps to reduce production costs. In addition, since growth conditions can be controlled in hydroponic systems, the effect of seasonal changes is kept at a minimum level. In this way, continuous production is provided throughout the year, not seasonal production [17]. It provides an advantage, especially in the cultivation of leafy vegetables, such as lettuce, which grow quickly and have high commercial value. Initially, the installation and

maintenance costs of hydroponic systems may be high, but in the long term, this is offset by high productivity and low operating costs [5]. Lettuce is a widely grown plant in hydroponic systems. The main reason for this is its fast growth process, low maintenance costs, and high commercial value [7]. Lettuce cultivation in hydroponic systems provides more control over the plant than traditional soil-based cultivation. The nutrients needed by the plant are directly transferred to the plant through the root, and nutrient deficiencies can be quickly detected and necessary solutions can be applied [15]. More homogeneous growth is achieved in plants grown in hydroponic systems, the harvest period is shortened and high and continuous quality is achieved in the plants that have become products [17]. In addition to lettuce, other leafy vegetables and herbs can be grown with soilless agriculture [16]. The future of soilless agriculture has become increasingly important due to the rapid increase in the world population and the decrease in agricultural land. The problems brought by climate change increase the need for sustainable and water-saving agricultural methods. Technologies such as vertical farming, closed-circuit hydroponic systems, and automation increase the potential and efficiency of soilless agriculture [16].

Today, the use of technology in agriculture has been a great help to farmers in optimizing agricultural production, especially in times of climate problems, with its efficiency and resource use. The study conducted by Waheed et al. [18] emphasized that deep learning techniques have the potential to make more accurate determinations compared to traditional methods used in agriculture. The study presents early examples of how artificial intelligence models can be used in agricultural practices and reveals important findings about innovative approaches in the field [18]. In recent years, artificial intelligence, data analytics, and automation systems have been widely used in hydroponic systems [19]. Artificial intelligence-supported sensors instantly monitor the amount of water, nutrients, and light that plants need and analyze the data to provide a suitable environment for the plant. These data make automatic adjustments to ensure optimum growth conditions [20]. It prevents unnecessary resource use by optimizing the growth process of plants. At the same time, plant health is examined with Internet of Things (IoT) devices and artificial intelligence, and early detection of diseases and deficiencies and prevention of plant loss are provided [21]. Not only in these areas, robots and drones are increasingly used to automate other agricultural tasks. These technologies help to reduce costs and increase production levels by reducing labor [22]. These innovations supported by artificial intelligence in hydroponic systems not only increase efficiency but also create a sustainable production process. It makes agricultural processes more predictable and manageable [23]. In a review study, the use of AI in hydroponic systems was discussed as a basic tool for monitoring and optimizing plant growth. The study emphasized the use of various sensors, such as pH, electrical conductivity (EC), temperature, and light intensity. It was stated that the data obtained from these sensors was processed using machine learning algorithms and AI models so that plant development could be managed more precisely [24]. Another study focused on the integration of AI and IoT technologies. The research

shows the potential of these technologies to increase agricultural efficiency by optimizing water and nutrient use. It was emphasized that AI algorithms monitored plant growth processes in real time and automatically regulated environmental conditions. It has been stated that these technologies are a big step, especially for the efficient use of resources and sustainable agriculture. Addressing the challenges in sustainable food production due to the limitations and resource inefficiencies of traditional soil farming, the study investigated the smart integration of hydroponic systems with IoT and artificial intelligence technologies. Through literature review and technology assessment, the use of IoT sensors and artificial intelligence-based control mechanisms in hydroponic agriculture was focused on. The results show that these smart systems optimize water and nutrient usage, improve plant growth, increase production efficiency, and enable sustainable automation in agricultural processes [25]. In the study, AI-based hydroponic systems were developed for lemon basil cultivation. In these systems, AI and IoT integration was used to monitor and control environmental factors. The study showed that AI can analyze data from sensors and cameras to monitor plant health and growth stages. The results revealed that AI has great potential to optimize plant growth performance [26]. Recent advances in hydroponics systems increasingly incorporate AI to increase efficiency, sustainability, and productivity [27]. A significant study explored the integration of smart technologies, including AI and IoT, in hydroponics. The authors highlighted how these technologies enable real-time monitoring and precise control of environmental factors, thereby optimizing crop growth and reducing resource usage. Their research highlighted the importance of data collection and remote cultivation facilitated by AI-enabled systems to ensure consistent and optimal conditions for plant growth [27]. Another significant study focused on the use of AI in aquaponics, combining hydroponics with aquaculture. The study examined various AI applications that addressed the complexities of maintaining a balanced ecosystem in aquaponics systems. The authors noted that AI can automate processes, such as nutrient management and system monitoring, thereby reducing human error and increasing the reliability of these systems. Despite the potential, the study also addressed challenges, such as energy efficiency and high costs associated with the implementation of these technologies [28]. AI and deep learning models are increasingly being used in various applications in agriculture, such as disease detection, yield prediction, and nutrient deficiency diagnosis. Deep learning models, such as convolutional neural networks (CNNs) have shown remarkable performance in image classification tasks [29]. Recent studies have demonstrated the effectiveness of CNNs in optimizing hydroponic systems through advanced image processing techniques [30]. For example, a study developed a hybrid CNN model specifically designed for early detection of tomato crop diseases in hydroponic environments. Integrating a starter module with a traditional CNN architecture, this model achieved an impressive accuracy of 99.17%, significantly enhancing the ability to monitor and maintain crop health in controlled environments [30]. Another study presented an adaptive deep learning model specifically designed for smart greenhouse environments. This model

used a one-dimensional CNN to analyze real-time data collected through IoT sensors and optimized the growth conditions of various crops, such as tomatoes and cucumbers. The system's sensitivity in controlling environmental factors, such as light, water, and temperature has been shown to be highly effective, with experimental results showing high accuracy rates in crop management, ultimately leading to improved yield and resource efficiency [31]. Studies on the detection of nutrient deficiencies in lettuce plants demonstrated how AI techniques, such as deep learning and CNNs can be used in agricultural applications [32]. For example, the study examined the effectiveness of CNNs in detecting nutrient deficiencies in plants. In this study, the model trained using labeled plant images was able to detect different nutrient deficiencies with high accuracy. The potential of AI to automate the detection process was highlighted by this study and is considered an important step in increasing productivity in agriculture [33]. Although the detection of plant nutrient deficiencies in this study was achieved with high accuracy, the generalization ability of the model over different environmental conditions and plant varieties has not yet been fully tested. Furthermore, further work on the computational efficiency and speed optimizations of the model is required for real-time applications.

Some of the You Only Look Once (YOLO) models have been frequently used in agricultural applications. One of these, the study conducted and included a different topic [34]. This was used for weed control. Here, the aim was to detect weeds in a wheat field. Here, YOLO versions from v3 to v5 were evaluated and comparisons were made between them. As a result of this study, the YOLOV5 model showed the best performance among other architectures. It was determined that artificial intelligence models closed a major gap in providing weed control. The proposed YOLOV5 model achieved approximately 77% success [34]. While it is usually used to detect nutrient deficiencies and diseases in plants, its use for weed detection has shown how much the needs in the agricultural field are and how many studies are being done. Another study, focused on using the YOLOV3 model to detect diseases in tomato plants. The model was trained on a dataset containing images of tomato leaves affected by various diseases. The YOLOV3 model showed high accuracy in identifying different types of diseases, making it a valuable tool for early diagnosis and treatment in agriculture [35]. In another study, combined YOLO-V8n with EfficientNet-v2s to create a model that can detect pest damage in hydroponic lettuce. The model successfully detected pests at different growth stages, making significant progress in automatic monitoring of hydroponic systems [36]. In this study, a lightweight version of YOLO called GVC-YOLO was developed for real-time detection of leaves damaged by cotton aphids. The model was designed to run on edge computing devices, addressing the challenges of processing power and resource constraints in the field. By reducing the complexity of the model and improving feature extraction capabilities, the researchers were able to deploy the model on Jetson Xavier NX and achieved impressive detection speeds and accuracy. This model is particularly important for large-scale pest control in cotton farming [37]. A study investigated the application of YOLOV5 to detect diseases in bell pepper plants. The model

was trained and validated using a comprehensive dataset of leaf images, and high levels of accuracy were achieved in identifying various disease symptoms. The study highlighted the importance of using deep learning models, such as YOLOV5 for early disease detection to prevent significant crop losses and improve yield quality [38]. This research specifically focused on using YOLOV3 for precision agriculture to identify and classify different plant diseases. The model was trained on a dataset of diverse plant images and successfully identified diseases with high accuracy. The study discussed the potential of YOLOV3 to improve the efficiency and sensitivity of disease detection in various crops, contributing to better resource management and crop protection strategies [39]. Several challenges remain in the application of AI models to nutrient deficiencies. Collecting and labeling such datasets is often a resource-intensive process and requires collaboration between agronomists and data scientists. Furthermore, while AI models can achieve high accuracy under controlled conditions, their performance may deteriorate in real-world scenarios where lighting and environmental factors vary. While publicly available datasets are available, they often reflect limited environmental conditions and fall short of real-world diversity. This limits the generalizability of AI models developed for early detection and nutrient management, especially in soilless agriculture for crops such as lettuce. Therefore, models with more comprehensive and flexible data sources need to be developed.

Although soilless farming systems today offer significant opportunities for sustainable and efficient production, there are some fundamental weaknesses that prevent the widespread use and optimum performance of these systems. In current hydroponic applications, especially in large-scale agriculture, there are deficiencies in fast and accurate monitoring of environmental variables and nutrient solution management. Traditional observation-based or manual intervention-based approaches are inadequate in detecting plant nutrient deficiencies at an early stage and in adapting to sudden changes in environmental conditions. In addition, current automation and artificial intelligence-based solutions are not widely applied in small and medium-sized enterprises, mostly due to the complexity and high costs of the systems. These weaknesses prevent the full potential of soilless farming from being fully utilized and limit the increase in yield and quality. Therefore, addressing the fundamental deficiencies of current systems, such as early detection, dynamic environmental adaptation, and cost-effective automation, is a critical requirement for soilless farming to become a sustainable and widespread solution.

In addition, current soilless farming systems have limited flexibility in plant nutrient solution management and adaptation to environmental variables. Failure to detect nutrient deficiencies and stress factors at an early stage, especially in fast-growing crops, such as lettuce, leads to yield losses and quality reductions. Furthermore, current automation solutions are not suitable for widespread use due to their high cost and complexity. These weaknesses prevent soilless agriculture from realizing its full potential and are a critical obstacle to sustainable and economical production.

In recent years, the increasing popularity and developments in the field of AI in agriculture have aroused curiosity.

When the studies in the literature are examined in general, it is seen that performance evaluations are mostly made with CNN-based models or old versions of YOLO and improvements are made on these models. However, it is noteworthy that in the current literature, there is no comprehensive performance comparison between the latest YOLO versions and that the datasets, especially consisting of agricultural images, are limited. The main purpose of this study is to eliminate these deficiencies, to compare both existing and proposed advanced models, and to analyze the real-time performances of current object recognition algorithms with an original dataset. In this context, YOLOV11 and YOLOV12 models developed with original architectural structures have been proposed. These models are designed to go beyond the classical YOLO architecture and provide a more optimized layer structure, faster inference time, and high accuracy with limited data. The YOLOV12 model is positioned at the focal point of the study with its current architectural features and hybrid module design.

The dataset used in the study is an original visual dataset designed to detect nutrient deficiencies in lettuce plants grown under a hydroponic system. This dataset; it consists of six classes: healthy, calcium deficiency, magnesium deficiency, nitrogen deficiency, phosphorus deficiency, and potassium deficiency. These nutrients are the elements that are essential for the healthy growth of the plant. These images obtained in real-world conditions made it possible to test the developed models not only in laboratory environments but also in real-time agricultural applications. At the same time, since hydroponic systems are also used, it is aimed for these nutrients to reach the plant completely. If there is a deficiency, it is aimed to be detected quickly and to help in taking the necessary actions. And at the same time, to determine the YOLO model that gives the best performance. In this way, it is aimed to contribute to the literature by keeping both the dataset and the developed model up-to-date.

The problems addressed in this study and the contributions of the study are presented below:

- The lettuce dataset used in the study was originally created in a hydroponic environment; it has a multiclass structure that includes healthy and different nutritional deficiencies. All images were labeled manually in detail.
- Two new model architectures named YOLOV11 and YOLOV12 were developed. These models are unique with their optimized layer structure, deeper architectural combinations, and improved connection strategies, unlike classical YOLO structures.
- The developed models are designed to work in real time even with limited hardware resources. This increases the potential for practical use in time-sensitive application areas, such as agriculture.
- In a way rarely done in the literature, the most up-to-date YOLO models (YOLOV8, YOLO-NAS, etc.) and the proposed models were compared extensively; they were analyzed in terms of accuracy (mean average precision [mAP]), speed frame per second (FPS), and resource consumption.

- The study provides a basis for fast and reliable decision support systems that can be used directly in the field of agricultural technologies. In this respect, it offers a model proposal that provides not only academic but also sectoral and societal benefits.
- The dataset was expanded with various data augmentation techniques to enable the model to adapt to different conditions. In this way, high accuracy was achieved despite the limited image, and the risk of overfitting was reduced. The findings obtained are an important reference that encourages the use of artificial intelligence for the purpose of increasing productivity in the agricultural sector and sheds light on future studies.

The importance and motivation of the study stem from the fact that the use of AI and deep learning technologies should be increased to monitor plant health and optimize nutrient management in hydroponic systems. This study demonstrates how deep learning-based solutions can be applied in the agricultural sector to increase efficiency, optimize resource use, and support sustainable agricultural methods. In particular, it reveals that YOLO architectures are a powerful tool for rapid and accurate detection of plant nutrient deficiencies in agriculture.

Today, the use of automation and artificial intelligence in agriculture has great potential. In particular, rapid detection of nutrient deficiencies allows producers to adjust nutrient solutions more efficiently. This study demonstrates the potential of deep learning models in providing data analysis and automation in agriculture, while also making a significant contribution to the literature. While there are studies in the existing literature on YOLO models mostly on object recognition and distribution, this study shows how deep learning techniques can be used on a much more specific and critical issue, such as nutrient deficiencies in agriculture.

In conclusion, this study highlights the high success of the YOLOV12 model used to detect nutrient deficiencies in lettuce plants in hydroponic systems. This success shows that the use of artificial intelligence to increase productivity in agriculture will increase and that agricultural technologies will develop further in the future. In addition, such deep learning-based solutions are considered an important step in developing sustainable agricultural practices and using resources more efficiently. In the future, testing these models on larger data sets and different plant species will contribute to opening up wider application areas in agriculture.

In conclusion, this study demonstrates the effectiveness of deep learning methods in the detection of nutrient deficiencies encountered in lettuce cultivation in hydroponic systems and offers a new perspective for the development of artificial intelligence applications in agriculture. This approach creates important opportunities to increase agricultural productivity and support sustainable agricultural practices.

2. Materials and Methods

In this study, YOLOV12 and YOLOV11 models were selected. In determining the most appropriate model, existing models were evaluated both by applying them and by comparing their

TABLE 1: Parameters for training.

Parameters	Setup
Epochs	100
Momentum	0.937
Batch size	40
Input image size	224 × 224
Workers	8
Amp	True
Wiegth decay	0.0005
learning_rate	0.001
Optimizer	Adam

performance results. During the training process, we trained the models using the Ultralytics library [38]. We performed the training process with GPUs on Google Colab. The GPU we used is Nvidia L4, and the CUDA core is 12.2. The GPU works with 22.5 GB RAM. The same parameters were used in all trainings.

As seen in Table 1, training was performed with the same parameters in all models during the training phase. The number of epochs was 100, the image size was 224 × 224, and the batch size value was 40, which would be the most optimum we obtained in the tests. Eight workers and an amp were used. As written in Table 1, hyperparameters for the YOLO model significantly affect the accuracy and overall performance of the model. Choosing these parameters correctly is critical to ensure that the model works quickly, accurately, and efficiently. Hyperparameter adjustments are usually done experimentally, and the role of each parameter in the success of the model is carefully analyzed [39, 40].

Learning rate is an important parameter that determines how much the model will change its weights at each iteration. A very high learning rate can cause the model to converge before reaching the optimum solution, while a very low learning rate can make the training process very slow and cause it to get stuck in a local minimum. In this study, the learning rate was selected as 0.001. This value is widely used in most deep learning models and generally performs well [41]. The model can be learnt faster and more stably by starting with a lower learning rate at the beginning and decreasing the learning rate (learning rate decay) when necessary.

Batch size determines how much data the network will be updated with at each training step. Small batch size provides more precise weight updates but increases computational cost and uses more memory [42]. Batch size provides faster training but may reduce the overall accuracy of the model. In this study, batch size is determined as 40. This value provides a good balance in terms of both computational efficiency and model accuracy in most deep learning models.

Epoch refers to the number of times the model will pass the training dataset. Generally, more epochs allow the model to learn more; however, the risk of overfitting also increases [43]. In this study, the training process is determined as 100 epochs. This value is usually enough time to ensure that the model learns sufficiently. During training, the validation set was used to prevent over-learning, and the early stopping technique

was also applied. In this way, the model was stopped when it showed signs of over-learning in the early stages.

The optimization algorithm determines how the model will update its weights [44]. In this study, the Adam optimization algorithm was used. Adam is an accelerated optimization method that adjusts the learning rate separately for each parameter [45]. This method was preferred because it showed high performance, especially in large datasets and complex models.

The momentum parameter is a hyperparameter that determines the speed and direction of weight updates during the training process of the model [46]. Momentum accumulates gradient information from previous iterations, allowing the model to reach the optimal solution point faster and more stably. The momentum parameter plays a critical role in increasing the accuracy of the model and making the training process more stable. In this study, a commonly used value of 0.937 was selected for the momentum parameter [47]. This value was preferred because it shows good performance in most deep learning models. When the momentum value is determined as 0.937, the gradients from previous iterations are strongly effective, while more resistance is shown against fluctuations in the training process. Thus, the model reaches the correct solution faster, while noise and oscillations in training are reduced. The intersection over union (IoU) threshold is used to determine whether a prediction is correct in object detection. This threshold expresses the overlap rate between the estimated bounding box and the actual bounding box. An IoU threshold of 0.5 is generally accepted as a value for correct detection because it means that the detected object must overlap with the real object by at least 50%. The IoU threshold of 0.5 was chosen because it provides a sufficient overlap rate for detections to be considered correct. A higher threshold (e.g., 0.75) requires more sensitive detection, while a lower threshold (e.g., 0.3) may yield more false positives. Therefore, 0.5 provides a reasonable balance between accuracy and coverage and is widely used in most detection tasks.

2.1. Dataset. The dataset consists of lettuce grown in a hydroponic system. It consists of six classes (Figure 1). These classes are healthy, calcium deficiency, magnesium deficiency, nitrogen deficiency, phosphorus deficiency, and potassium deficiency. Each of these is a plant's essential nutrient and their deficiencies negatively affect the plant's development. This prolongs the growth period that we want to keep under control in hydroponic systems. The images are 1086. Preprocessing and data augmentation methods were applied to the images for training. Our data was provided in partnership with a facility in the region. Plants in growth were divided into groups. Then, the nutrient that was missing was added to the nutrient solution of each plant, and the color change and other reactions that this plant exhibited in the absence of nutrients were provided, and then the data set was created by obtaining their images. This data set is original and is actively used in development activities in a facility.

Preprocessing steps to be performed before classifying nutrient deficiencies in plant leaves with the YOLO architecture eliminate distortions in the images and increase the



FIGURE 1: Sample image from the dataset.

accuracy of the model. First, the dimensions of the images are resized according to a certain standard, and the pixel values are normalized between 0 and 1. These preprocessing steps improve the image quality, shorten the processing time, and increase the prediction accuracy. Preprocessing techniques used in this study include image enhancement, noise injection, cropping, brightness, zoom, and contrast changes. In addition, more samples were added to the dataset using various data augmentation techniques (rotation and translation). Finally, correctly labeled data is prepared, divided into training, validation, and test sets. These steps are necessary for the model to perform correct classification and increase the success of the training process. The images were first labeled for object detection. Then, as a preprocessing, they were resized and brought to the input value of 224×224 . In the image enlargement process, horizontal and vertical rotations, as well as 90-degree rotations were made clockwise, counterclockwise and upside down. As a result of these operations, the total number of images was 2978. As seen in Figure 2, more than one labeling was made in an image. For classes where the number of images is 2978, the distribution in the number of labeled data, that is, how many labels were made for that class in all images. Table 2 shows the distribution of labeled data. 80% of the data was reserved for training and the rest for validation. After these processes, the resulting dataset had a total of 2978 images. While there were 2870 images for training, 478 images were sufficient for validation.

In the study, the effects of data augmentation techniques were investigated in five different scenarios in order to increase image classification performance (Table 3). In each scenario, it was observed how the performance levels of the proposed models changed with various augmentation methods applied to the image data. Thanks to this analysis, it was revealed that problems, such as overlearning and memorization, which are frequently encountered in classification problems, can be

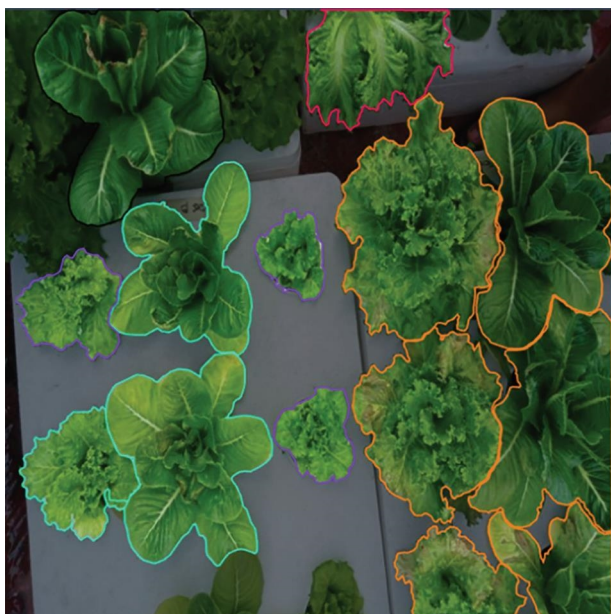


FIGURE 2: Example labeled data.

TABLE 2: Number of labels.

Class	Train	Validation
Healthy	2650	517
Calcium deficiency	2655	496
Phosphorus deficiency	2578	498
Nitrogen deficiency	2560	498
Potassium deficiency	2287	484
Magnesium deficiency	1900	398

TABLE 3: Number of labels images for all data.

Scenario	Total image
1	2978
2	4455
3	5692
4	7734
5	10,315

significantly reduced through data augmentation techniques. In Scenario 1, only the original 2978 images were used. In other scenarios, the number of images was increased, and the dataset was expanded with GAN (generative adversarial networks) and traditional data augmentation methods [48]. Detailed information about the number of images obtained with these methods is presented in Table 3. In the data augmentation process, synthetic images were produced by applying various operations such as rotation, translation, scaling, contrast enhancement, image size change, cropping, blurring, and adding noise. In addition, with the GAN-based approach, images belonging to six different classes in the dataset were synthetically recreated, class imbalance was eliminated and the overall dataset was made more balanced. Data augmentation techniques were applied to all classes at the same rate, and synthetic images

were labeled appropriately. Thus, comparative analysis could be performed by preserving the distribution between classes in each scenario. In Table 3, the total number of images for each scenario is given in detail, and their effects on model performance are presented comparatively in the following sections.

2.2. YOLO. YOLO works by reducing the object detection problem to a regression problem. This method scans the input image once and estimates object classes and locations simultaneously [49]. The basic methodology consists of the following steps:

Grid-based segmentation: YOLO divides the input image into a grid system of a certain size. Each grid cell analyzes whether there is an object in the relevant region. For example, to examine color changes in plant leaves, each cell analyzes pixel intensities and color distributions.

Bounding box estimation: YOLO estimates the bounding box coordinates and class scores for the detected objects simultaneously. In the case of nutrient deficiency detection, these bounding boxes mark the leaf regions where color changes occur.

Determining class probabilities: By estimating the class probabilities for each bounding box, the model can identify different nutrient deficiencies (e.g., yellowing due to nitrogen deficiency or pale coloration in new leaves due to iron deficiency).

Using anchor boxes: Anchor boxes are used to detect color changes of different shapes and sizes more accurately. This is especially useful for capturing color changes that occur in large areas or small localized regions.

In the methodology for detecting nutrient deficiencies, it is important to label the dataset we prepare. Then, the training of the model is usually done by fine-tuning the pretrained YOLO network with the transfer learning method. The trained model is optimized to understand the color differences in the leaves by examining each pixel of the images. YOLO has a CNN-based architecture and extracts features, such as color intensity, patterns, and contrast. For example: color contrast between leaf veins and the surrounding area. The trained model analyzes each region of the leaf, and the detected color changes are marked with bounding boxes. Prediction results are presented with class scores.

2.2.1. YOLOV11. YOLOV11 is one of the latest and most advanced versions of the YOLO series developed by Ultralytics. Built on YOLOV8, this model includes significant architectural innovations to deliver higher accuracy, speed, and efficiency. YOLOV11 is available in five different configurations as follows: nano (*n*), small (*s*), medium (*m*), large (*l*), and extra-large (*x*) to adapt to different tasks and hardware needs. Each of these structures has been optimized considering processing power and memory limitations [50].

One of the most notable innovations is the C2PSA (cross-stage partial self-attention) module, which combines cross-stage partial (CSP) connections with the self-attention mechanism, allowing both low-level and high-level features to interact effectively [50]. This structure provides a particularly significant performance increase in the detection of small or close-together objects.

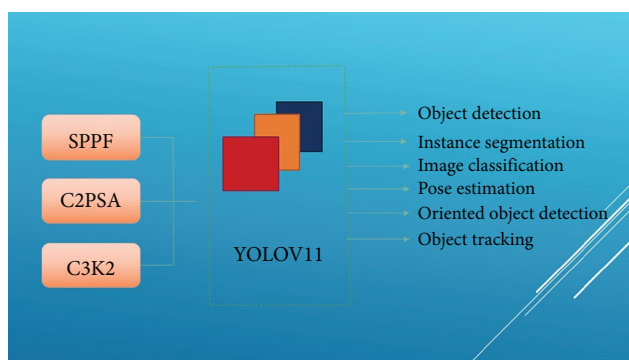


FIGURE 3: YOLOV11.

In addition, the C3k2 block used in the YOLOV11 architecture offers a more efficient approach compared to traditional CSP structures (Figure 3). This structure reduces the computational load by using only two convolution layers, while minimizing information loss. Thus, the model becomes a faster and less resource-consuming structure with a lower number of parameters.

The prominent technical advantages of YOLOV11 can be listed as follows [51]:

- OTFE (optimized target-aware feature extraction): Thanks to this module, the feature extraction phase of the model is optimized sensitively to the target object, which provides more stable and faster learning.
- DCIP (dual-channel interaction pathway): Developed specifically to increase contextual awareness, this structure takes into account both local and global context through two-channel interaction. Thus, the model can detect objects more precisely.
- Parameter-efficiency balance: YOLOV11 enables real-time operation on low-equipment devices (e.g., mobile systems, edge devices) by achieving higher accuracy with fewer parameters. This makes it suitable for a wide range of uses from industrial applications to autonomous systems.
- Efficient training process: Thanks to the advanced architecture, the model reaches higher accuracy in a shorter time and the training process becomes more stable. This enables rapid prototyping in both academic and industrial environments. Thanks to these advanced modules and architectural optimizations, YOLOV11 not only offers higher performance compared to previous versions but also represents a significant step forward in terms of flexibility of use, processing efficiency, and hardware compatibility.

2.2.2. YOLOV12. YOLOV12 is one of the latest generation object detection models developed by Ultralytics and offers significant improvements by pushing the performance and efficiency limits of previous versions. The model aims to provide both high accuracy and fast inference in real-time applications [52].

YOLOV12 can be configured with five different scales as follows: nano, small, medium, large, and extra-large. These scales are optimized for different application and hardware

needs. Thus, it can be used effectively on both mobile devices and powerful servers.

From a technical perspective, the most basic innovations of YOLOV12 are as follows [53]:

- Advanced multiscale feature fusion: YOLOV12 increases the detection accuracy of small objects and difficult-to-detect objects by more effectively combining feature maps obtained from different layers. Thus, it provides high performance even in complex scenes.
- Dynamic layer structure and flexible network topology: The model has the ability to dynamically adjust the network depth and width according to the structure of the dataset and task requirements. In this way, flexibility and efficiency are provided in both training and inference stages.
- Optimized version of C3-convolution blocks: C3 blocks used in YOLOV12 have been revised to provide more computing capacity with fewer parameters. This brings improvements in both speed and accuracy.
- Innovative loss functions for accurate classification and localization: The model uses optimized hybrid loss functions to minimize classification and localization errors. This means more accurate box estimates and fewer false detections.
- Advanced regularization techniques: YOLOV12 includes new regularization methods and dropout variations to prevent overfitting. This increases overall performance and increases model generalization capacity.
- Lightweight and efficient feature extraction layer: This layer, optimized specifically for use on edge devices, provides high-quality feature extraction with low latency.

Thanks to these technical improvements, YOLOV12 not only offers higher accuracy and speed, but also provides a strong foundation for a wide range of applications in real-time object detection. The model is suitable for use in many areas, from autonomous vehicles to video surveillance systems, from mobile applications to industrial automation.

2.3. Methodology. In this study, images of lettuce plants were collected, as shown in Figure 4 and subjected to a comprehensive preprocessing process to increase the accuracy of the proposed models. The dataset was manually labeled into six different classes, including healthy plants and calcium, magnesium, nitrogen, phosphorus, and potassium deficiencies.

In the preprocessing stage, images that were very similar to each other were eliminated to increase the data quality, all images were scaled to appropriate resolutions and sizes, and normalization processes were applied to adapt to the model inputs. After this preparation process, the dataset was divided into 80% and 20% for training, and testing, respectively.

In the training stage, both YOLOV11 and YOLOV12 models designed with original architectural structures were used. Hyperparameter settings were carefully determined and improved with optimization techniques to maximize model performance. As a result, both models were trained and comparative performance analyzes were performed.

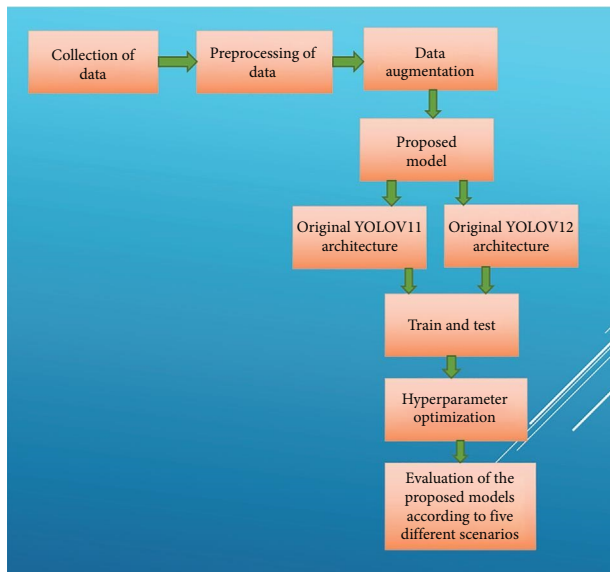


FIGURE 4: System architecture.

In this study, two new attention mechanisms were developed and integrated to increase performance based on the YOLOV11 architecture. First, the multiscale features obtained from the spine and neck layers are dynamically reweighted at the channel and spatial level with the hierarchical feature attention module (HFAM), thus increasing the model's sensitivity, especially for small objects and complex scenes. Second, the contextual awareness of the model is strengthened with the dynamic context enhancement layer (DCEL) in hybrid datasets containing different lighting and image types. DCEL, which uses the combination of channel and spatial attention mechanisms, increases the generalization ability and detection accuracy against environmental changes. These unique modifications make significant contributions to the literature by increasing the robustness and effectiveness of YOLOV11-based systems in real-time applications. In this study, two new attention-based modules are added to the YOLOV12 architecture to increase the classification accuracy. First, the HFAM, which dynamically reweights feature maps at different scales with channel and spatial attention mechanisms, is integrated into the neck layer. Thus, the detection sensitivity, especially for small objects and complex scenes, is significantly improved. Second, the DCEL, which was developed to increase the context awareness of the model against heterogeneous datasets and variable environmental conditions, was included in the neck output. DCEL, which uses SE and CBAM-based attention mechanisms, increases the robustness and generalization ability against environmental changes. Thanks to these unique modules, YOLOV12 offers an innovative contribution to the literature by showing superior performance in terms of both speed and accuracy in real-time applications.

3. Experimental Study

We have done our training so far. First, let's look at what our evaluation parameters are.

Train/box_loss: This evaluation metric represents the box regression loss, which calculates the difference between the

actual and estimated box sizes and coordinates. The accuracy in the estimated boxes is shown with smaller box_loss [54].

Train/cls_loss: This evaluation metric represents the classification loss, which calculates the difference between the actual and estimated class probabilities. The lower cls_loss, the better the model predicts the classes.

Train/df_l_loss: This evaluation metric is a new feature used for YOLO models. It shows the convolution layers that are ignored and corrupted to increase the object recognition accuracy. As df_l_loss decreases, the changes in the model object definition are clearer and it shows that the data samples that are difficult to predict are better predicted.

Val/box_loss: It represents the loss in box changes during validation.

Val/cls_loss: It represents the classification loss during validation.

Val/df_l_loss: It represents the focal loss distributed during validation across epochs.

Metrics/precision(B): Shows false positive predictions during training.

Metrics/recall(B): Shows true positive predictions during training.

Metrics/mAP50(B): Shows average precision for IoU 50 across epochs during validation. Increasing the mAP50 evaluation metric indicates an improvement in the model's performance in object recognition.

Metrics/mAP50–95(B): It shows the average precision from 50 to 95 during validation in IoU over epochs. An increase in the mAP50–95 evaluation metric indicates an improvement in the performance of the model in object recognition.

F1 Confidence curve: F1 - Score is calculated by the harmonic average of precision and recall metrics. F1 Confidence curve expresses the F1 score according to different confidence thresholds. As the F1 score increases, the performance increases and the confidence threshold with the highest F1 score is the most suitable threshold value in classification accuracy.

Precision-recall curve: Precision-recall curve shows the balance between precision and recall at different threshold values. Precision is the ratio of true positive predictions to total positive predictions.

Precision–confidence curve: This curve determines the precision values according to different confidence levels. It shows how the precision criterion changes according to the confidence level. As the confidence threshold increases, precision increases.

Recall-confidence curve: Determines the recall criterion according to different security levels. Shows how the recall criterion changes according to the confidence level. As the confidence threshold increases, the recall decreases.

We evaluated the models according to the above parameters. Now let's look at the results of the models one by one.

4. Results and Discussion

In Table 4, when the mAP50 and mAP50–95 performances of YOLOV11 and YOLOV12 models are compared for all

TABLE 4: Results of the proposed models for five scenarios.

Dataset/YOLO architectures	YOLOV12		YOLO11	
	mAP50	mAP50-95	mAP50	mAP50-95
Scenario 1	0.984	0.957	0.975	0.944
Scenario 2	0.962	0.929	0.957	0.916
Scenario 3	0.948	0.912	0.931	0.902
Scenario 4	0.924	0.887	0.903	0.873
Scenario 5	0.902	0.871	0.893	0.867

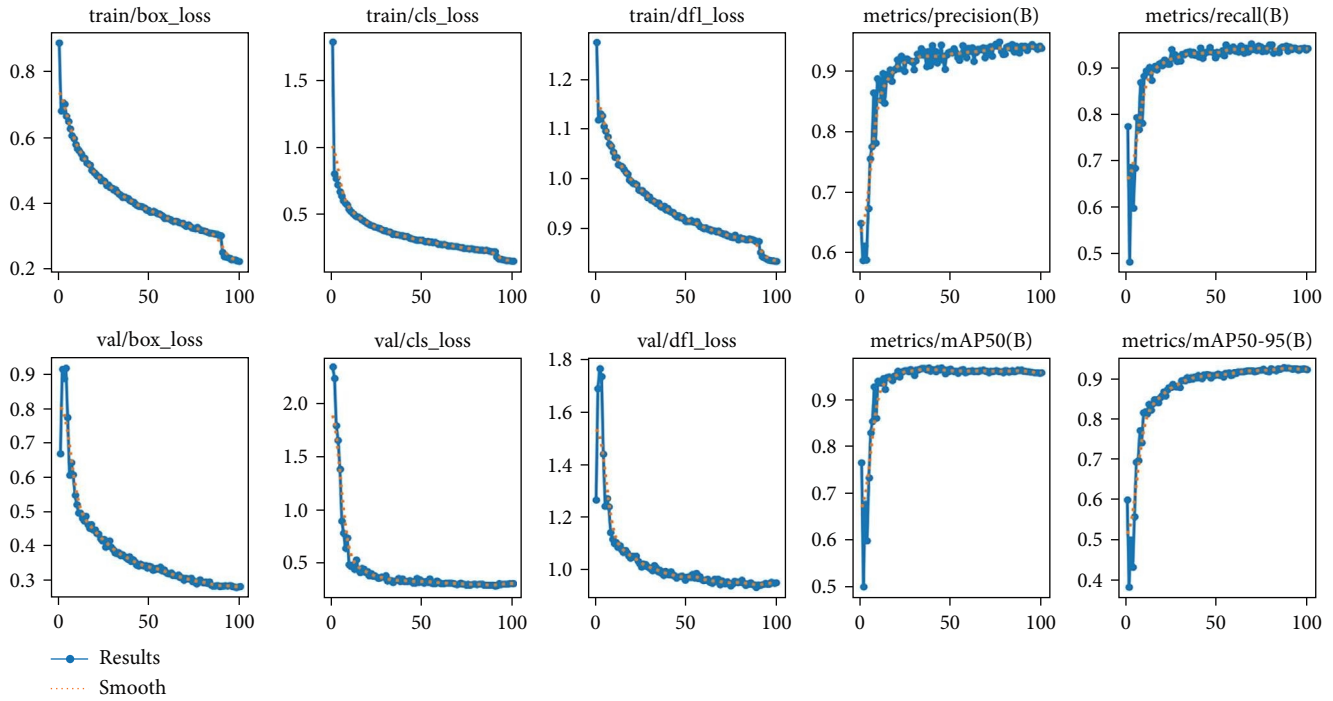


FIGURE 5: YOLOV12 result graphs.

scenarios and classes, it is seen that YOLOV12 provides significantly higher success. The main reason for this difference is the more advanced and optimized architectural structure of YOLOV12 and the improved hyperparameter settings. In this way, the model achieves higher accuracy and precision in both feature extraction and classification. In addition, the training strategy improvements and design innovations made in YOLOV12 have contributed to the superior performance of the model in important metrics, such as mAP50 and mAP50-95 by increasing its generalization ability on different datasets and complex scenarios. These improvements show that YOLOV12 is a significant step forward compared to the previous version in terms of both accuracy and consistency.

As a result of comparisons made on five different scenarios, it has been seen that YOLOV12 architecture achieves higher accuracy rates compared to YOLOV11 architecture. The basis of this superiority is the optimized architectural design and improved hyperparameter settings of YOLOV12, as well as the integrated attention-based modules. In particular, the dynamic reweighting of multiscale features at the channel and spatial level with HFAM has increased the detection

success of the model on small and complex objects. In addition, the model has become more robust to different environments and data conditions and contextual awareness has been improved thanks to the DCEL. These innovative and unique modules have enabled YOLOV12 to show more robust and consistent performance in real-world applications, thus achieving significant improvements compared to the previous version.

4.1. YOLOV12. The graphs of training, validation loss, and metrics (mAP50 and mAP50-95) for YOLOV12 are shown in Figure 5. While train/box_loss, train/cls_loss, train/dfl_loss show the training losses, train/box_loss, train/cls_loss, train/dfl_loss show the validation losses. Training and validation losses show a continuous decrease, which shows that the YOLOV12 model is an effective model. In addition, metrics/precision(B) and metrics/recall(B) evaluation criteria for training and validation express their positive dynamics and indicate that the model is improved. It is shown that the fluctuations caused by the model decrease by increasing the data with various data augmentation techniques. It can be seen that the

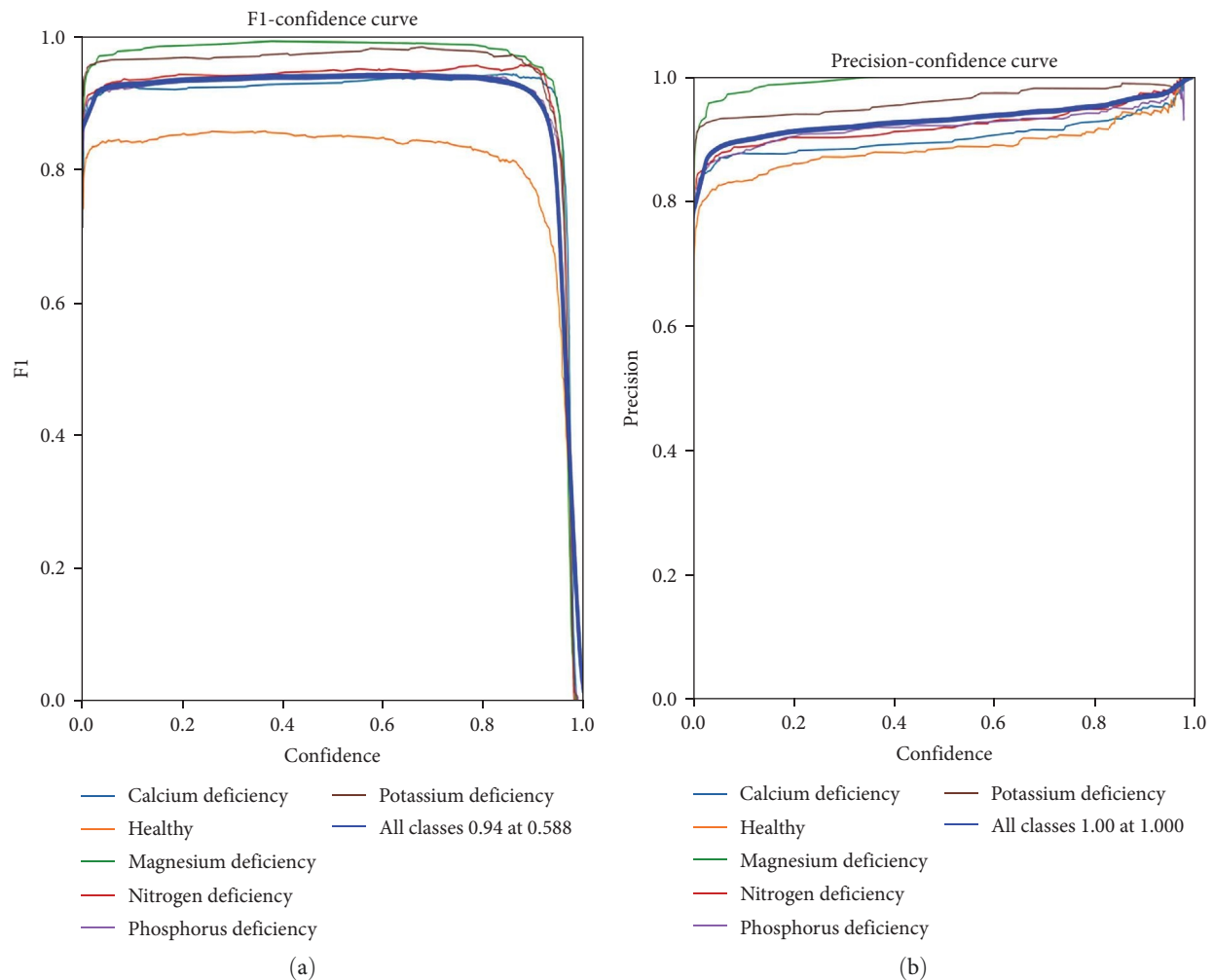


FIGURE 6: (a) YOLOV12 F1-confidence curve and (b) YOLOV12 precision-confidence curve.

training and validation losses slowly decrease throughout the training process. Training continues until a certain stopping criterion. The fluctuations in the box loss decrease over time throughout training and validation. The classification loss shows a continuous decrease and reaches a constant state over time. It also shows that there is no anomaly in the df_loss evaluation metric for training and validation, and the success of predicting difficult examples is seen to increase.

Figure 6(a) shows the F1-score curve for the YOLOV12 model for all classes (six classes). The F1-Confidence curve represents the balance between precision and recall. The high F1-score shows that the YOLOV12 model performed well in terms of precision and recall. The class with the highest F1-score is determined by the high prediction confidence. When we look at the high prediction confidence of all classes for YOLOV12, the “Magnesium Deficiency” class has the highest F1-score (approaching 1). The overall model performance for all classes is shown by the thick blue line. YOLOV12 has approximately 94% sensitivity at a confidence threshold of 0.588 for all classes. It shows that it peaks at 0.588 at a confidence level of 0.94. This peak is quite significant for YOLOV12 and leads to the highest combined model performance measure

in determining nutrient deficiency in lettuce. Figure 6(b) shows the Precision–confidence graph for the YOLOV12 model for all classes (six classes). This graph expresses the change in sensitivity at different confidence intervals. It shows that the average precision (mAP) is 100% in all classes at the IoU threshold of 1.0. This indicates that there is perfect sensitivity at a high confidence threshold. According to the graph, the “Magnesium Deficiency” class is the most perfect class in terms of sensitivity at different confidence intervals.

Figure 7(a) shows the precision-recall graph for YOLOV12 model for all classes (six classes). Precision recall–curve graph shows the balance between precision and recall for different thresholds. Average Precision (mAP) for YOLOV12 model is 96.3% for all classes at IoU threshold of 0.5. “Magnesium Deficiency” class is the best class at IoU threshold in the precision-recall graph.

Figure 7(b) shows the recall-confidence graph for the YOLOV12 model for all classes (six classes). The recall-confidence graph shows the recall at different confidence thresholds. The lowest confidence threshold (0.00) has a 98% recall in all classes as shown in Figure 7. This shows that the YOLOV12 model correctly predicts the true positive in

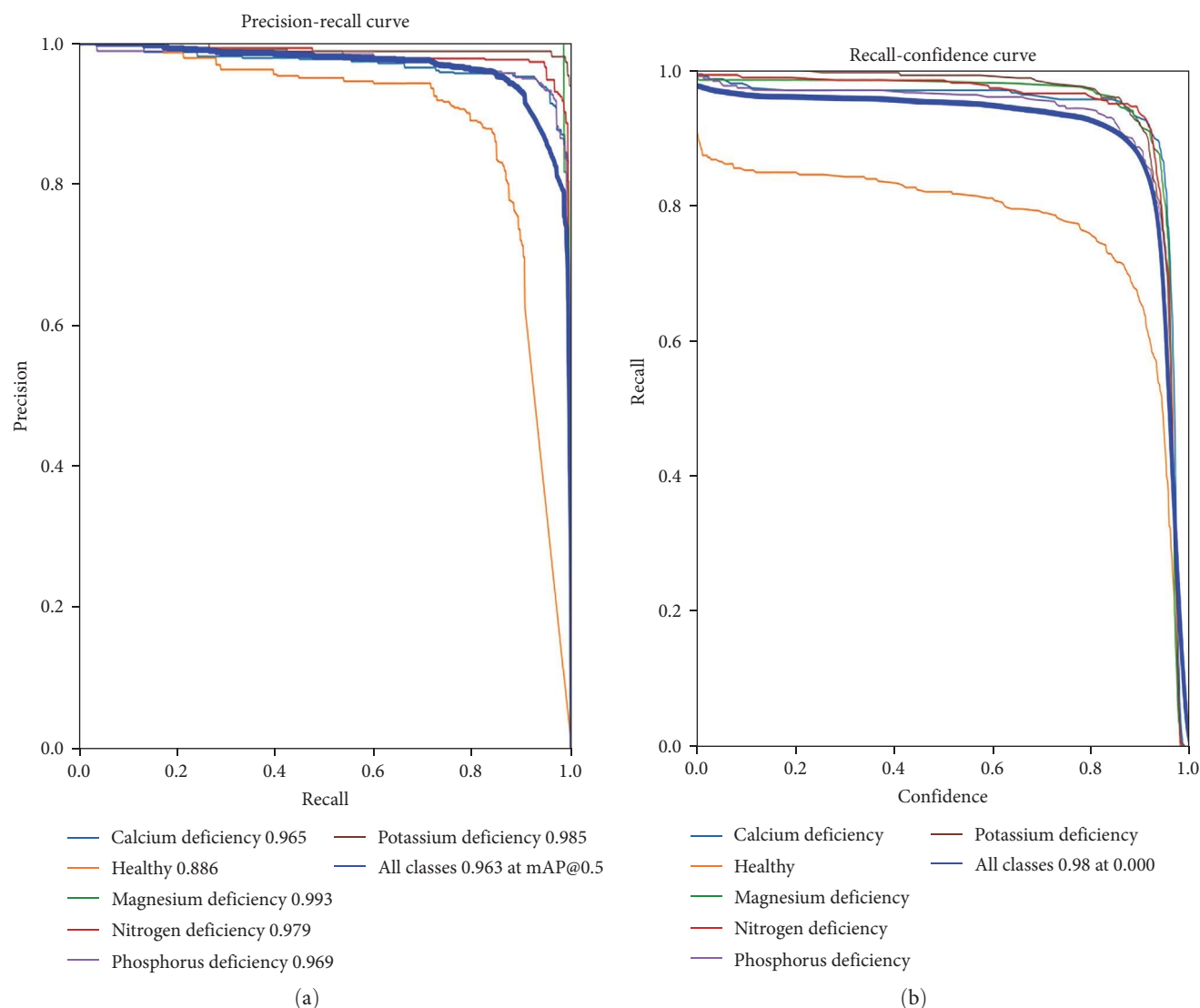


FIGURE 7: (a) YOLOV12 precision-recall curve and (b) YOLOV12 recall-confidence curve.

determining the mineral deficiency in lettuce nutrient deficiency. According to the graph, the “Magnesium Deficiency” class is the class with the best recall at different confidence intervals.

The confusion matrices of the YOLOV12 model, presented in Figure 8, reveal that the model detected the “Potassium Deficiency” class with higher accuracy compared to other classes. According to the normalized confusion matrix, 99% of the samples belonging to this class were correctly classified. The “Potassium Deficiency” class stands out as the class with the highest accuracy. This is likely due to the small number of samples belonging to this class, allowing the model to learn patterns related to this class more clearly. On the other hand, the fewest samples belonged to the “Healthy” class, and only 81% of the samples belonging to this class were correctly classified. This suggests that the limited number of samples and the similarity between classes can negatively impact model performance.

4.2. YOLOV11. The training and validation results of YOLOV11 models using the augmented lettuce nutrient deficiency dataset are shown in Figure 9 train/box_loss, train/cls_loss, train/dfl_loss represent the box loss, class loss, and distribution focus loss, respectively train/box_loss, train/cls_loss, train/dfl_loss represent the bounding box loss, class loss, and distribution focus loss, respectively, during the validation process metrics/mAP50 (B), metrics/recall (B) represent the evaluation criteria. mAP is a frequently used metric to measure the performance of computer vision and deep learning models in object detection and detection. It provides information about the location and classification success of deep learning models. mAP is found by calculating the average of the average sensitivity measurement of all classes in the used models. It takes a value between zero and one. mAP50, B is used to distinguish segmentation metrics and shows the average sensitivity at the IOU = 0.50. According to Figure 9, for any selected image sample, the most optimum values are calculated as

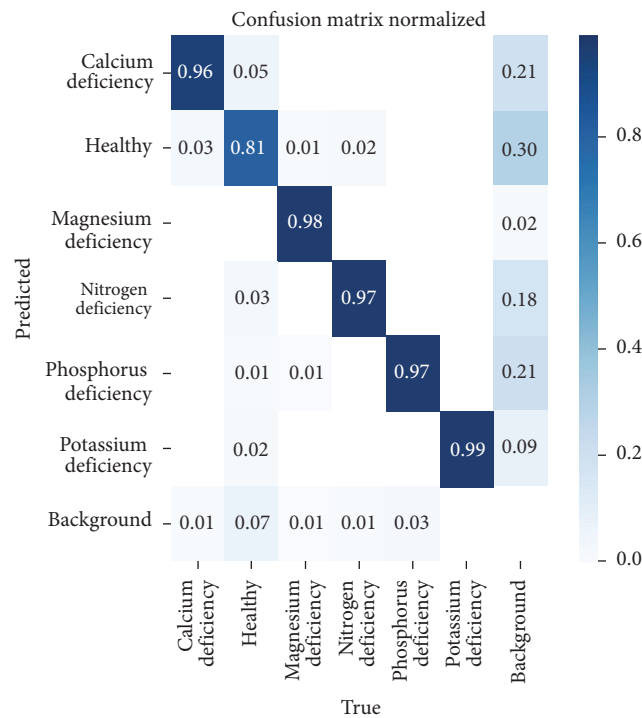


FIGURE 8: YOLOV12 normalized confusion matrix.

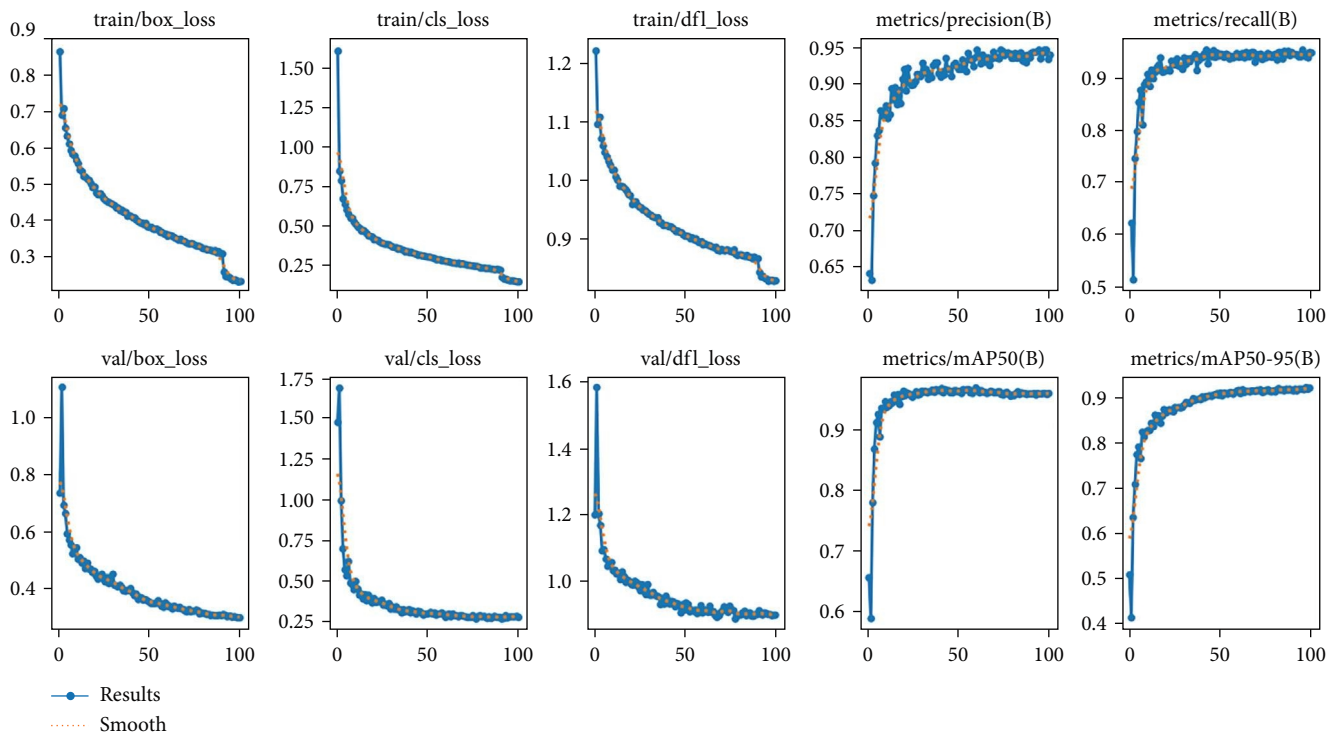


FIGURE 9: YOLOV11 result graph.

approximately 0.93 precision and 0.92 recall at the 99th epoch. The mAP50–95 evaluation metric shows the average precision calculated at various IoU thresholds ranging from 0.50 to 0.95. According to Figure 9, it is observed that for any selected image

sample, the most optimum values are approximately 0.94 precision and 0.90 recall at the 99th epoch. Figure 10 shows the F1-score curve for the YOLOV11 model for all classes. Considering the high prediction

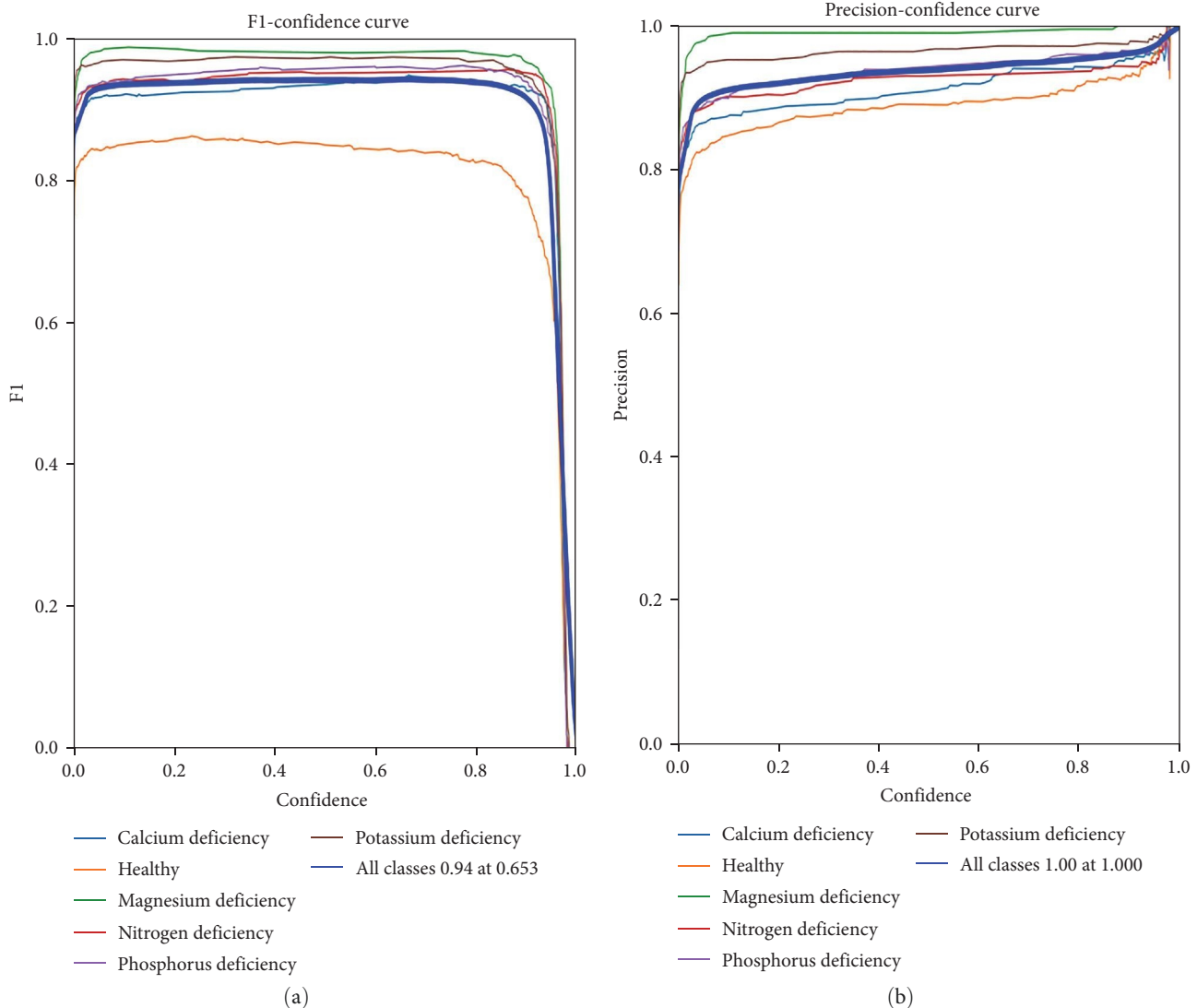


FIGURE 10: (a) YOLOV11 F1-curve and (b) YOLOV11 precision-curve.

confidence of all classes for YOLOV11, the “Magnesium Deficiency” class has the highest F1-score. YOLOV11 has approximately 94% sensitivity at the confidence threshold of 0.653 for all classes. It shows that it peaks at 0.653 at the confidence level of 0.94. This peak point is quite significant for YOLOV11 and leads to the highest combined model performance measure in determining nutrient deficiency in lettuce.

Figure 10(b) shows the Precision–Confidence graph for the YOLOV11 model for all classes. It shows that the Average Precision (mAP) is 100% in all classes at the IoU threshold of 1.0. This indicates that there is perfect precision at a high confidence threshold. According to the graph, the “Magnesium Deficiency” class is the most perfect class in terms of sensitivity at different confidence intervals.

Figure 11(a) shows the Precision–Recall graph for YOLOV11 model for all classes. Precision Recall – Curve graph shows the balance between precision and recall for different thresholds. For YOLOV11 model, mAP shows 96.2% in all

classes at IoU threshold of 0.5. “Magnesium Deficiency” class is the best class in the precision–recall graph at IoU threshold.

Figure 11(b) shows the Recall–Confidence graph for the YOLOV11 model for all classes. The Recall–Confidence graph shows the recall at different confidence thresholds. The lowest confidence threshold (0.00) has a 98% recall in all classes as shown in Figure 11. This shows that the YOLOV11 model correctly predicts the true positive in determining the mineral deficiency in lettuce nutrient deficiency. According to the graph, the “Potassium Deficiency” class is the class with the best recall at different confidence intervals.

The confusion matrices of the YOLOV11 model presented in Figure 12 show that the model detected the “Magnesium Deficiency” and “Nitrogen Deficiency” classes with higher accuracy compared to the other classes. According to the normalized confusion matrix in Figure 12, 98% of the samples belonging to these two classes were correctly classified. The samples with the highest accuracy rate belong to these classes.

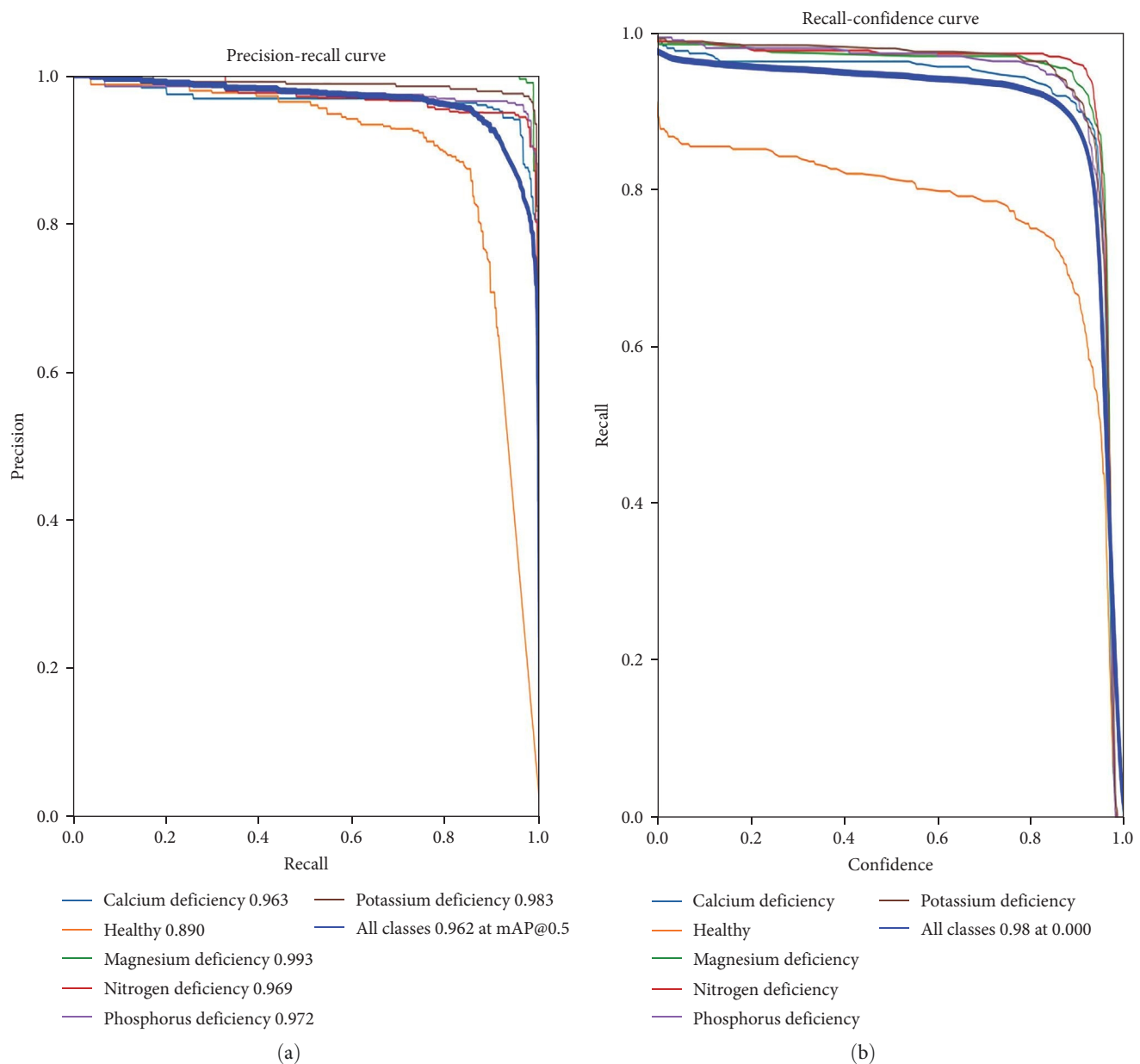


FIGURE 11: (a) YOLOV11 precision-recall curve and (b) YOLOV11 recall-confidence curve.

This result suggests that the model may have learned the patterns specific to these classes more easily because these deficiencies appear with more distinct visual symptoms in the lettuce plant. At the same time, the balanced and distinct sample distribution in these classes may have increased the classification success.

4.3. Hyperparameter Tuning and Results for Different Epoch Values of the Same Batch for Scenario 5. Table 5 includes the evaluations of the YOLOV11 version, which showed the highest success performance under Scenario 5, based on the mAP50 and mAP50–95 metrics obtained for different epoch numbers. The analysis results reveal that there is a significant increase in success rates in parallel with the increase in the number of epochs. This situation ensures that the model is exposed to

the data more in each iteration process and learns the patterns more deeply, and thus its generalization capacity improves significantly. Indeed, increasing the number of epochs in the training process contributes to the more accurate optimization of the model's parameters, a more balanced progress of the learning process and ultimately an increase in the level of accuracy. In this context, the ability of the model working with more epochs to both minimize its errors and grasp complex patterns increases, which is directly reflected in the overall performance. Therefore, the findings confirm that longer training cycles, which encourage deep learning of the model, offer a decisive advantage in terms of performance.

Table 6 presents the evaluation results of the YOLOV12 version, which achieved the highest accuracy rates under Scenario 5, according to the mAP50 and mAP50–95 performance

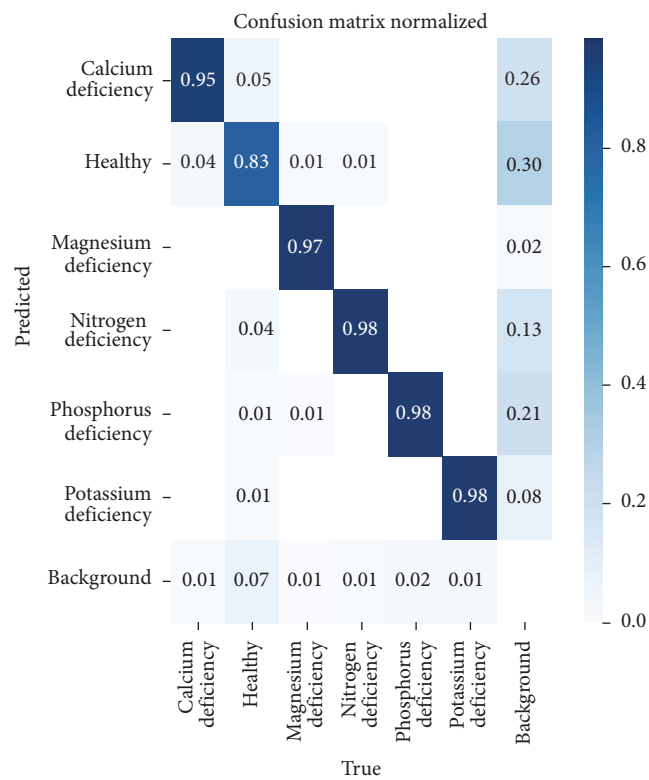


FIGURE 12: Normalized YOLOV11 confusion matrix. Performance evaluation metrics for the experimental setup.

TABLE 5: YOLOV11 results for Scenario 5 according to different epoch values.

Epoch	120		150		200	
	mAP50	mAP50–95	mAP50	mAP50–95	mAP50	mAP50–95
	0.901	0.873	0.920	0.893	0.933	0.907

TABLE 6: YOLOV12 results for Scenario 5.

Epoch	120		150		200	
	mAP50	mAP50–95	mAP50	mAP50–95	mAP50	mAP50–95
	0.928	0.893	0.937	0.906	0.947	0.914

metrics calculated over different epoch numbers. The findings show that the increase in the epoch number has a direct positive effect on the accuracy performance of the model. In particular, the increase in the success rate of the YOLOV12 version indicates that the model strengthens its ability to learn richer patterns and systematically reduce its errors with more training cycles. Higher epoch values allow the model to adjust its parameters more precisely during the training process, balance the learning curve, and develop more robust generalization ability. This process contributed to YOLOV12 achieving more consistent and higher success levels in critical metrics such as mAP50 and mAP50–95. As a result, extending the training period increased the learning depth of the model, thus significantly improving both the accuracy rates and the overall performance.

4.4. Hyperparameter Tuning and Same Epoch Values for Different Batch Results for Scenario 5. Table 7 presents the performance results of the YOLOV11 version in line with the mAP50 and mAP50–95 metrics calculated according to different batch size values within the scope of Scenario 5. According to the analysis findings, a significant increase in the success rates of the model was observed as the batch size value decreased. This can be associated with the fact that the model can make more frequent weight updates in the training processes performed using smaller batch sizes. Small batch values allow the model to make more precise adjustments in the learning process by taking into account more diverse examples in each update step. In addition, this prevents the model from overgeneralizing and helps to create a more balanced and sensitive learning process. Especially in small data sets or scenarios

TABLE 7: YOLOV11 results for Scenario 5.

Batch_size	4		8		16	
	mAP50	mAP50–95	mAP50	mAP50–95	mAP50	mAP50–95
	0.922	0.889	0.913	0.874	0.904	0.870

TABLE 8: YOLOV12 results for Scenario 5 according to different batch_size values.

Batch_size	4		8		16	
	mAP50	mAP50–95	mAP50	mAP50–95	mAP50	mAP50–95
	0.938	0.913	0.924	0.897	0.917	0.885

with complex patterns, low batch size values support the model to learn more refined and can increase accuracy rates. As a result, the fact that the YOLOV11 version achieves higher success rates with low batch size values indicates that the training process is optimized with more frequent and effective updates, which positively affects the model performance.

Table 8 presents the mAP50 and mAP50–95 performance metrics obtained as a result of training the YOLOV12 version with different batch size values within the scope of Scenario 5. The obtained results show that there is a significant increase in the model's accuracy rates with the decrease in the batch size value. This increase is based on the fact that the model can update the parameters more frequently in each training cycle thanks to being trained with smaller datasets. Low batch size values allow the model to update its weights more frequently, making the learning process more dynamic and sensitive. Thus, the model learns complex patterns more precisely and gains better generalization ability by preventing overgeneralization. This situation plays an effective role in increasing the accuracy metrics, especially in deep artificial neural network architectures, such as YOLOV12. As a result, the training processes performed with low batch size increased the learning sensitivity of YOLOV12 and provided a noticeable performance increase in both mAP50 and mAP50–95 values.

4.5. Results of All Scenarios According to k -Fold and Different Evaluation Metrics. The results in Table 9 show that the YOLOV11 model achieved higher success in cross-validation with $k = 10$ than with $k = 5$. This is because the model works with more training data each time with $k = 10$ and can generalize better thanks to more frequent testing. A model that is sensitive to details such as YOLOV11 has shown an increase in accuracy metrics by learning patterns more deeply with this additional data. In addition, this method has produced more balanced and reliable results by reducing the risk of overfitting. Training the YOLOV11 model with more folds such as $k = 10$ increases the generalization ability by providing the model with more diverse data samples and reduces the risk of overfitting. This situation reveals both the scalability of the model and its flexibility against different data distributions; thus, it clearly shows that YOLOV11 has a strong and reliable architecture.

According to the results in Table 10, the YOLOV11 model in Scenario 1 showed the best performance in terms of precision, recall, and F1-score compared to the other scenarios. Although the amount of data in Scenario 1 was less than the

TABLE 9: YOLOV11 k -fold results for all scenarios.

Scenario	$k = 5$	$k = 10$
1	0.964	0.981
2	0.952	0.965
3	0.947	0.953
4	0.924	0.937
5	0.901	0.914

TABLE 10: Results of YOLOV11 model according to other evaluation metrics of test data.

Scenario	Precision	Recall	F1-score
1	0.943	0.929	0.914
2	0.937	0.924	0.907
3	0.927	0.921	0.894
4	0.903	0.911	0.884
5	0.895	0.884	0.879

others, the model detected nutrient deficiencies in the lettuce plant with clearer and more distinct features. This situation allowed the model to focus more on important patterns with the decrease in data diversity and reduced the confusion between classes. In addition, the fact that the precision values were higher than the recall and F1-score in all scenarios shows that the model prioritizes keeping false positives low but may miss some true positives. This is due to the model being more selective in order to minimize false alarms. The fact that Scenario 1 showed high success even with less data indicates that the model effectively learnt important and distinctive features. The performance differences observed in all scenarios were due to the model prioritizing minimizing false positives. This approach caused the model to be more selective when making decisions, thus causing the precision values to be higher than the recall values.

The results shown in Table 11 show the k -fold cross-validation performance of the YOLOV12 model. In particular, the accuracy rates of the model at $k = 10$ are significantly increased compared to $k = 5$. This improvement is due to the model's wider exposure to different data subsets by using more training data with smaller validation sets during the training process. Thus, YOLOV12's generalization ability is

TABLE 11: YOLOV12 k -fold results for all scenarios.

Scenario	$k = 5$	$k = 10$
1	0.976	0.992
2	0.958	0.973
3	0.942	0.967
4	0.917	0.943
5	0.898	0.937

TABLE 12: Results of YOLOV12 model according to other evaluation metrics of test data.

Scenario	Precision	Recall	F1-score
1	0.967	0.948	0.928
2	0.958	0.933	0.917
3	0.942	0.927	0.911
4	0.924	0.921	0.906
5	0.919	0.905	0.894

TABLE 13: Comparison of different model performance metrics of YOLOV11 for all scenarios.

Scenario	Inference time (ms)	FPS (frame per second)	FLOPs (G)
1	232	18.1	9.1
2	243	18.3	9.2
3	249	18.4	9.4
4	257	18.6	9.5
5	267	18.8	9.7

strengthened, and it has become more robust to different data distributions. In addition, the superior performance of the model on large and complex datasets supports the scalability of its architecture and its flexibility against data diversity. These features increase the reliability of the model, especially in real-world applications.

As seen in Table 12, while testing the YOLOV12 model in different scenarios, Scenario 1 showed the highest performance in terms of precision, recall, and F1-score on lettuce. The main reason for this is that the dataset in Scenario 1 contains nutrient deficiency symptoms with clearer and more consistent visual features, thus facilitating the model to make correct classifications. In this way, the model both reduced false positives and increased its overall success by detecting deficiencies more accurately.

4.6. Comparison of Different Model Performance Metrics of All Scenarios Across All YOLO Architectures. In the tests conducted with scenarios with different data densities on the proposed YOLOV11 architecture for the detection of nutrient deficiencies in lettuce plants, the metrics related to the processing efficiency of the model were compared. When Table 13 is examined, it is seen that the inference time (ms), FPS, and FLOP (G) values in Scenario 5 are significantly increased compared to Scenario 1. This difference is directly related to the higher data volume processed in Scenario 5. The increase in

TABLE 14: Comparison of different model performance metrics of YOLOV12 for all scenarios.

Scenario	Inference time (ms)	FPS (frame per second)	FLOPs (G)
1	245	19.3	11.2
2	267	19.4	11.3
3	278	19.7	11.4
4	280	19.8	11.5
5	293	19.9	11.6

the amount of data required the model to process more visual information. This both increased the computational intensity (FLOP) and extended the inference time, causing a relative decrease in FPS. Although such variables are important, especially in real-time systems, it was observed that YOLOV11 exhibited consistent and scalable performance even under increased load. These findings reveal that the model works stably even with high data volumes and can be optimized for real-time applications.

Table 14 presents the inference time (ms), FLOP (G), and FPS values for the performance of the YOLOV12 model in different scenarios. According to the data, these metrics have increased significantly in Scenario 5 compared to Scenario 1. This is due to the fact that the model requires more intensive computing due to the larger amount of data used in Scenario 5. The increased data volume has extended the inference time of the model, increased the computational load (FLOP) and caused a decrease in the FPS value. These findings show that YOLOV12 can perform strong feature extraction in large datasets but requires higher computational resources accordingly. In summary, while the model maintains its high accuracy potential, optimization is needed according to the scenario and data size in real-time applications.

5. Conclusion and Discussion

In this study, performance comparisons of all architectures and model structures used to detect nutrient deficiencies in lettuce plants are presented in Table 15 for Scenario 1. Performance metrics, such as precision, recall, mAP50, and mAP50–95 were calculated using the originally developed YOLOV11 and YOLOV12 models. The YOLOV12 model stands out as the most up-to-date and advanced version among the architectures in the YOLO family. Therefore, it exhibited higher accuracy and overall performance in all classes compared to previous YOLO versions. This is primarily because YOLOV12 is an upgraded version that addresses the structural limitations of previous versions and implements various optimizations.

When all classes were evaluated, it was observed that the “Magnesium Deficiency” class was detected with the highest accuracy by all YOLO architectures. In experiments conducted using the augmented dataset, the YOLOV12 model demonstrated superior performance, achieving mAP50 of 0.995 and mAP50–95 of 0.981 across all classes. When the individual performances of the YOLO architectures on the test samples were examined, an overall balanced classification performance was achieved across the six classes. However, the lowest success

TABLE 15: Performance results of YOLO architectures trained on lettuce nutrient deficiency.

Model	Class	Box precision	Box recall	mAP50	mAP50–95
YOLOV11	All	0.962	0.939	0.975	0.944
	Calcium deficiency	0.963	0.956	0.927	0.912
	Healthy	0.890	0.792	0.890	0.807
	Magnesium deficiency	0.993	0.970	0.988	0.974
	Nitrogen deficiency	0.969	0.973	0.954	0.928
	Phosphorus deficiency	0.972	0.968	0.930	0.919
	Potassium deficiency	0.983	0.974	0.974	0.973
YOLOV12	All	0.963	0.949	0.984	0.957
	Calcium deficiency	0.965	0.970	0.941	0.938
	Healthy	0.886	0.812	0.902	0.814
	Magnesium deficiency	0.993	0.981	0.995	0.981
	Nitrogen deficiency	0.979	0.977	0.965	0.937
	Phosphorus deficiency	0.969	0.961	0.956	0.925
	Potassium deficiency	0.985	0.991	0.979	0.976

rates were reported for classifying instances belonging to the “Healthy” class across all architectures.

This may be due to the visual characteristics of instances belonging to the “Healthy” class being similar to those of classes with nutritional deficiencies. In particular, images where symptoms are not yet apparent or where there are significant differences are more likely to be confused with other classes by the models. Furthermore, the low intraclass variation in instances belonging to the “Healthy” class may have limited the model’s ability to generalize to this class.

In the study, the performances of all architectures and models used to determine nutrient deficiency in lettuce are shown in Table 3 for all classes and individual classes. Precision, recall, mAP50, and mAP50–95 metrics were calculated for each YOLO version. Since YOLOV12 version is the newest and most advanced version among YOLO architectures, it is more successful in all classes compared to previous architectures. Because it is an improved and developed version of previous YOLO versions. When all classes were evaluated, the “Magnesium Deficiency” class was detected more accurately in all architectures. The reason for this is that this class has fewer examples. YOLOV12 achieved 0.984 mAP50 and 0.957 mAP50–95 in the increased dataset in all classes. When the performances of YOLO architectures were compared separately on the test examples, a balanced trend was observed among the six classes. It was observed that the lowest results were obtained in YOLOV11 with 0.975 mAP50 and 0.944 mAP50–95 in all classes. However, it gave the lowest results in all YOLO architectures in the classification of samples belonging to the “Healthy” class.

There are visual similarities between “healthy” leaves and some types of nutrient deficiencies. For example, a slight initial color loss or structural changes may be misclassified as “healthy” by the model. There may be more uncertainty in the definition of the “healthy” category than other categories. For example, since certain small natural variations of a healthy leaf are perceived by the model as a sign of nutrient deficiencies, the healthy class has achieved a lower accuracy value.

Table 16 compares this study with recent studies in the literature. The methods used in recent studies, their success rates, and their advantages and disadvantages are given. Overall 98.4% success was achieved with YOLOV12 with the original multiclass dataset. When the study results are evaluated, the results and findings obtained in this study are quite valuable.

The performance of the YOLO architectures used in this study was evaluated in detecting nutrient deficiencies in lettuce plants. The analyses performed on the YOLOV11 and YOLOV12 models show that the YOLOV12 model stands out with its high accuracy and mAP results. Each of the models achieved successful results in identifying different classes of nutrient deficiencies, but it was observed that the YOLOV12 model performed better than the other models with 98.4% mAP. In addition, the reduction of model fluctuations thanks to various data augmentation techniques and the improved structure of the model increased the detection accuracy.

In this study, the YOLOV11 and YOLOV12 models, developed to detect nutrient deficiencies in lettuce plants grown in hydroponic systems, were extensively tested across five different dataset size scenarios. Performance evaluations were conducted using multiple metrics, such as precision, recall, mAP50, and mAP50–95, with the YOLOV12 model, in particular, achieving an average accuracy of over 98% across all scenarios. This high performance strongly demonstrates the model’s potential to identify nutrient deficiencies early and accurately, minimize yield losses due to deficiencies, and optimize nutrient management. While superior performance was observed in the “Magnesium Deficiency” class, the classification difficulties experienced in the “Healthy” class highlight the limitations in identifying visual similarities and early-stage symptoms. These findings are important for developing more sensitive and efficient production strategies in agriculture.

Discussion of the results in light of the literature supports the role of artificial intelligence as an indispensable tool for optimizing nutrient management in agriculture and better understanding plant functions. For example, studies published in the Journal Plant Physiology [66] emphasized the

TABLE 16: Comparison of the proposed model with studies in the literature.

Study	Dataset	Method and accuracy (%)	Advantages and disadvantages
Rahadiyan et al. [55]	Created in Sleman, Yogyakarta, Indonesia	MLP %90.55 CNN %89.70	The proposed method is more successful than other methods. However, determining and monitoring agricultural environmental conditions is very important in plant health. Larger data sets are needed to determine the causes of nutrient deficiencies
Sikati and Nouaze [56]	NPK dataset	YOLOV8%93	It becomes more difficult to integrate smart agriculture applications in more complex systems
Taha et al. [57]	Taken from plants photographed at different times of the day with a PowerShot SX720 HS, Canon Inc. camera in Tokyo, Japan	ResNet18%92.1 SVM %86.1 KNN %84.5 decision tree %79.8	In real-time applications, disease detection in lettuce plants requires new images and more training. It is also inflexible
Li et al. [58]	Public database consisting of 1921 images	Improved YOLOV8%91.40	Disease detection in plants was performed by real-time disease detection
Khalid et al. [59]	Collected from agricultural university, Faisalabad city, Pakistan	YOLOV5%93	Further analysis can be performed on larger datasets
Jose et al. [60]	IPNI dataset	ANN %88.27	Advanced deep learning models can deliver better results
Xinming and Hong [61]	Plant village dataset	YOLOV4%78.60 YOLOV7%82.20	Environmental conditions, plant image resolution, and scale of plant leaf spread and infection stage may affect the success of the models
Vo et al. [62]	FruitDetectionv3 image dataset	YOLOV9%88.20	As a result of the study, the quality, productivity and sustainability of plants are very important for advances in agriculture
Rajamohan and Latha [63]	Few plants were photographed in agricultural fields in Kerala and Tamil Nadu	YOLOV5 93%	This study was conducted with very little data
Lu et al. [64]	Rice leaf data from Halahai farm in Meiris district, Qiqihar city, Heilongjiang province	Transfer learning algorithm 95.9%	Optimization algorithms can be used to optimize the hyperparameters
Firmando et al. [65]	Public dataset rice leaf images	InceptionV3 %90.00	More comprehensive analyses can be performed on larger datasets
Our study	On images augmented with data augmentation techniques	YOLOV12%98.4	With multiclass application, nutritional deficiencies are detected faster and more accurately

importance of artificial intelligence in the early detection of plant stresses and the optimization of nutrient balances. Similarly, some recent articles demonstrate that using multiple performance metrics, such as precision and recall, is critical for accurate and reliable results. The limitations of the systems proposed in these articles regarding generalizability and real-world applications are paralleled in our study by the relatively small dataset and potential for adaptation to real-world conditions [67, 68].

In this study, the performance of the YOLOV11 and YOLOV12 models, developed for the detection of nutrient

deficiencies in lettuce plants grown in hydroponic systems, was comprehensively evaluated across five scenarios with varying data sizes. Multiple metrics, including precision, recall, mAP50, and mAP50–95, were used as performance metrics, and the results demonstrated that the YOLOV12 model demonstrated superior performance with an average accuracy of over 98% across all scenarios. This demonstrates the effectiveness of deep learning-based approaches in optimizing nutrient management to reduce yield losses due to deficiencies. The model's particularly successful results in the "Magnesium Deficiency" class can be considered a significant advancement in

accurately identifying such specific nutrient deficiencies. In contrast, the classification difficulties observed in the “Healthy” class stem from the visual similarity of early-stage symptoms and low intraclass variation.

The performance differences between YOLO models are closely related to the data augmentation techniques used, the depth of the model and the computational requirements. While lighter model YOLOV11 can be trained quickly using fewer resources, deeper models, such as YOLOV12 have achieved higher accuracy rates due to advanced data augmentation techniques and training with larger datasets. However, it should be noted that these models require more computational resources and training time.

Although the higher accuracy provided by deeper models is especially useful in critical applications, such as detecting nutrient deficiencies in plants, it can create practical difficulties for small-scale farms or operations with limited resources. On the other hand, lighter models that can perform fast detection may pose risks in some cases due to the decrease in accuracy. Therefore, model selection should be made carefully depending on the application requirements and resources.

The performance differences of YOLO models can be explained by the effects of data augmentation techniques, model architecture, and computational resources. Future studies can focus on testing models on larger datasets, applying them to different plant species, and optimizing computational requirements. Such optimizations will enable wider use of automation and real-time detection applications in agriculture.

The results show that deep learning-based YOLO models can be used in agricultural applications and are effective in quickly and accurately detecting nutrient deficiencies in lettuce plants. Especially in agriculture, automation and artificial intelligence applications have great potential in increasing product efficiency and optimizing resource use. In this context, it is envisaged that YOLO architectures will play an important role in solving various problems in the field of agriculture in the future.

It makes a significant contribution to the literature by examining the usability of deep learning-based YOLO models in agricultural applications. In particular, it focuses on nutrient deficiency detection in lettuce plants and offers innovative solutions for nutrient management and plant health in hydroponic systems. While studies on YOLO models in the existing literature mostly focus on object detection and classification, this study reveals the potential of deep learning models in a specific area, such as nutrient deficiency detection in agriculture. The results obtained guide future studies on the use of artificial intelligence to increase efficiency in the agricultural sector.

As a result, this study has shown that deep learning models, especially YOLO architectures, can be used effectively in agricultural applications. In the analyses conducted to detect nutrient deficiencies in lettuce plants in hydroponic systems, the YOLOV12 model achieved the most successful results. These results reveal the potential of artificial intelligence in providing data analysis and automation in agricultural processes, while providing important clues on how the technology can be used for sustainable agriculture. Future studies can contribute to the

further development of agricultural technologies by testing these models on larger data sets and different plant species.

The results of this study can be applied not only to lettuce plants but also to different plant species. The usability of YOLO models in detecting nutrient deficiencies in different plant species has great potential in terms of increasing productivity and optimizing resource management in agriculture. For example, it may be possible to detect deficiencies in different vegetable species, such as tomato, pepper, or potato with similar methodologies. Considering the different nutrient requirements and disease symptoms of different plant species, the models may need to be trained separately for each plant species.

In addition, although this study is limited to hydroponic systems, the applicability of a similar approach in traditional soil farming systems should be emphasized. Determination of nutrient deficiencies in soil environments may vary depending on different soil conditions, water regulations, and environmental factors. Therefore, environmental and local factors should be taken into account for the applicability of YOLO-based detection systems in different agricultural environments. For example, environmental factors, such as soil pH, moisture content, and sunlight can change the appearance of nutrient deficiencies, so these parameters should also be taken into account in the training of the model.

Future research suggests that YOLO models should be tested in different environmental conditions. The environments where plants grow (e.g., greenhouses, outdoor farming, or indoor hydroponic systems) can vary with different light conditions, humidity, temperature, and other environmental factors. These conditions can affect the differences in the images of the plants, which can change the accuracy of the model. For example, in low light conditions, the images of the plants may be of lower quality, which can negatively affect the detection accuracy of the model. Similarly, in outdoor conditions, the visual signs of nutrient deficiencies in plants may differ compared to greenhouse conditions. Therefore, it is important to train with more data to increase the model's ability to generalize to environmental variations.

It should be noted that YOLO-based object detection models are not limited to detecting only nutrient deficiencies in agriculture but also offer great potential for early detection of pests, diseases and other stress factors. Diseases and pests in plants can be identified by visual symptoms, and YOLO models can also detect such stress factors. For example, problems such as fungal infections, bacterial diseases, or insect pests can manifest themselves in different ways on leaves or fruits. Therefore, YOLO models can be used to create a comprehensive disease and pest management system in agriculture.

In addition, these models have potential applications in soil and water quality monitoring systems. High levels of chemical fertilizer use in soil can cause nutrient deficiencies in some plant species. This can be associated with the visual deterioration detected by the model. It has also been observed that plant health is negatively affected in cases where water quality is low. The use of YOLO in such environmental monitoring systems can contribute to the development of environmentally friendly approaches in agriculture.

It is expected that artificial intelligence-based systems, especially deep learning methods, will find wider application in agriculture. Such technologies will form the basis for the introduction of automated monitoring and intervention systems in agriculture. Object detection algorithms, such as YOLO models are likely to become widespread in areas such as automatic disease detection, irrigation management, fertilization strategies, and general plant health monitoring in agriculture. In this way, farmers will be able to make faster and more accurate decisions with less human intervention, and productivity will be increased. It will also be possible to reduce resource use and minimize environmental impacts.

This study has shown that YOLO-based models show high performance in detecting nutrient deficiencies in lettuce plants. The findings show that this technology can be effectively used in agriculture, especially in plant health monitoring and management systems. Future research suggests that YOLO models should be tested with larger data sets across different plant species and environmental conditions, and the model should be optimized to adapt to environmental variation. Furthermore, considering that such AI-based systems offer great potential for sustainability and efficiency in agriculture, it is important to integrate the technology into wider agricultural areas.

Data Availability Statement

The data sharing is not applicable to this article.

Conflicts of Interest

The authors declare no conflicts of interest.

Author Contributions

“Cigdem Bakir and Ayberk Gezer contributed equally to this work.”

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